

Quantized Multimodal LLMs Say Less, Not More: Text-Side Grounding Loss and a Closed-Form Repair (Mid-Evaluation Report)

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<https://github.com/suryamanoj4/MLLM-quantization-research>

Abstract

Multimodal LLMs are routinely deployed with quantized weights, yet little is known about how quantization changes what they say about an image. The common expectation is that quantization damages the visual tokens, so the model leans on language priors and *hallucinates* objects that are not there. We test this on LLaVA-1.5-7B with five widely used quantizers and find the opposite. Quantized models do not invent more objects; they *omit* objects that are present. With 3-bit weights, captions mention up to 7.2 percentage points fewer of the objects annotated in the image, small objects are dropped five to seven times as often as large ones, and the standard benchmark, POPE, misses this entirely (F1 changes by at most 0.02). Quantizing the model only while it processes image tokens leaves captions intact; quantizing it only while it processes text tokens reproduces the full loss. The visual representations survive quantization; what fails is the language side that reads them out, which also attends 14–18% less to the image at 4 bits. Targeting this site, we introduce LoRAS-T, a low-rank correction of the decoder’s hidden states at text positions, fitted in closed form in 30 minutes without gradients. Adding only 34 MB, it brings 3-bit round-to-nearest from a 7.2-point loss to no significant difference from the full-precision model on every grounding measure we use.

1 Introduction

Multimodal LLMs (MLLMs) such as LLaVA (Liu et al., 2023, 2024a) answer questions about images and describe them. They are usually deployed quantized: storing the 7B language decoder in 16-bit floating point (FP16) takes 14 GB, and with 4-bit weights about 3.5 GB. Post-training quantization (PTQ) methods for MLLMs (Wang et al., 2024; Li et al., 2025; Yu et al., 2025; Xue et al., 2025; Zhong et al., 2026) are evaluated on general benchmarks, while methods against object hallucination (Leng

et al., 2024; Huang et al., 2024; Liu et al., 2025a) are designed and tested at full precision. Whether quantization changes *which objects a model reports seeing* has not been measured directly.

The natural hypothesis, and the starting point of this project, is *Lexical Fallback*: quantization noise corrupts the visual tokens, the decoder’s attention to them flattens, and the model falls back on language priors, inventing plausible but absent objects. If true, quantized MLLMs would need hallucination defences. We tested the hypothesis directly before building on it, and it fails in both its direction and its location.

What we find. Using validated re-implementations of the quantizers people deploy (NF4, AWQ, GPTQ, SmoothQuant and rotation), measures that separate invented objects from missed ones, and paired statistical tests on identical images, we establish four results on LLaVA-1.5-7B:

1. **Quantization causes omission, not hallucination.** Captions keep their length but name fewer of the objects that are present (up to −7.2 points at 3 bits) and invent no more, often fewer. POPE F1 stays within 0.02 of FP16 in all 16 deployed settings, so the standard evaluation misses the loss.
2. **Small objects are lost first.** At 3 bits, 28% of the small objects that FP16 mentions disappear from the captions, against 5% of large ones.
3. **The damage is on the language side.** Quantizing the decoder only while it processes text tokens reproduces the full loss; quantizing it only for image tokens causes none. Random weight noise of the same size also causes omission, and quantized models attend less to the image.

4. A correction at that site removes the loss.

LoRAS-T, a closed-form low-rank correction of text-position hidden states (34 MB), leaves 3-bit round-to-nearest (RTN) with no significant difference from FP16 on any grounding measure.

These results have practical consequences. Grounding evaluations of compressed MLLMs should report how many present objects are mentioned (coverage), not only hallucination rates, which improve when a model says less. Quantizers and repairs for MLLMs should target the decoder’s computation at text positions rather than protect the visual tokens. And a small closed-form correction recovers grounding at a fraction of the memory cost of an extra bit per weight.

2 Related Work

Post-training quantization. Round-to-nearest (RTN) with per-group scales is the baseline. GPTQ (Frantar et al., 2023) compensates each rounding error using second-order statistics of calibration inputs; AWQ (Lin et al., 2024) rescales salient channels by activation magnitude; LLM.int8() and SmoothQuant (Dettmers et al., 2022; Xiao et al., 2023) handle activation outliers; QuaRot and SpinQuant (Ashkboos et al., 2024; Liu et al., 2025b) rotate activations to enable 4-bit activations; OmniQuant (Shao et al., 2024) learns clipping and transforms; NF4 (Dettmers et al., 2023) is the 4-bit format of the bitsandbytes library. Massive activations (Sun et al., 2024) and attention sinks (Xiao et al., 2024) make low precision hard. Error compensation ranges from bias correction and adaptive rounding (Nagel et al., 2019, 2020) to low-rank terms added to quantized weights (Yao et al., 2023; Zhang et al., 2024, 2025a; Liu et al., 2024c; Li et al., 2024) or acting on activations (Scetbon and Hensman, 2024). Closest to our repair, Quantization without Tears (Fu et al., 2025) fits a per-block linear correction in closed form to match full-precision block outputs. All of these correct every token alike and target perplexity or accuracy; LoRAS-T is low-rank, acts only at text positions, and targets grounding.

Quantizing MLLMs. Q-VLM (Wang et al., 2024) searches block-wise quantizers; MBQ (Li et al., 2025) finds language tokens about $10\times$ more sensitive than vision tokens and reweights calibration by modality; SplitQ (Zhong et al., 2026)

separates modality-specific outliers; MQuant (Yu et al., 2025) enables static per-modality activation quantization; VLMQ and QIG (Xue et al., 2025; Xiang et al., 2026) weight calibration tokens by saliency; LUQ and QSLAW (Bhatnagar et al., 2026; Xie et al., 2024) assign bits per layer or learn scales. Das et al. (2026) quantize the vision encoder and language model separately, and Kang et al. (2026) find that lower reconstruction error does not reliably improve accuracy. All report general-benchmark accuracy; none measures which objects the model reports.

Object hallucination. POPE (Li et al., 2023b) asks yes/no questions about objects; CHAIR (Rohrbach et al., 2018) counts invented objects in captions; AMBER (Wang et al., 2023) adds object coverage (survey: Bai et al., 2024). Hallucination falls with shorter, less detailed captions (Yue et al., 2024), so CHAIR rewards saying less. Mitigations contrast predictions with and without the image (Leng et al., 2024), penalize over-trust in summary tokens (Huang et al., 2024), amplify image attention (Liu et al., 2024b), or steer hidden states (Liu et al., 2025a), as activation steering does for LLMs (Li et al., 2023a; Turner et al., 2023; Zou et al., 2023; Wu et al., 2024). Full-precision MLLMs already miss small details (Zhang et al., 2025b), many image tokens can be pruned (Chen et al., 2024), and some act as attention sinks (Kang et al., 2025).

Behaviour of compressed models. Compressed LLMs can keep their accuracy while changing many answers (Dutta et al., 2024); equal-size perturbations of different distribution harm LLMs differently (Gong et al., 2024); pruned LLMs hallucinate *less* in summarization (Chrysostomou et al., 2024). LVLM-Compress-Bench finds no major hallucination increase under 4-bit MLLM compression (Kundu et al., 2025). Two concurrent studies are closest to ours: GHOST-Q (Rehman and Shafique, 2026) shows that INT8 and NF4 VLMs keep benchmark scores while item-level POPE answers change, and Saha et al. (2026) show that 4-bit weights shift LLaVA’s yes/no margins before answers flip. To our knowledge, no prior work separates omission from hallucination under quantization, locates the damage at image or text positions, controls for error size, or repairs it.

3 Problem Setup

Question. An MLLM f with decoder weights W is quantized by a PTQ method Q to $\hat{W} = Q(W; \mathcal{D}_{\text{cal}})$, where $\mathcal{D}_{\text{cal}}$ is a small calibration set. On the same images and prompts, we compare the full-precision outputs f_W with the quantized outputs $f_{\hat{W}}$ and ask how the model’s *grounding*, the set of objects it reports as present, changes.

Measures. We measure grounding in two tasks. In *captioning*, the model describes an image; let G_c be the objects annotated in image c and M_c the objects its caption names (among the 80 COCO categories). Two complementary rates follow:

$$\text{CHAIR}_i = \frac{\sum_c |M_c \setminus G_c|}{\sum_c |M_c|},$$

$$\text{COV} = \frac{\sum_c |G_c \cap M_c|}{\sum_c |G_c|}.$$

CHAIR_i (Rohrbach et al., 2018) is the share of named objects that are *not* in the image (hallucination); coverage (COV) is the share of present objects that the caption names. A model that says less lowers both, so they must be read together. In *POPE* (Li et al., 2023b), the model answers “Is there a [object] in the image?” for present and absent objects. Saying “yes” to absent objects lowers *precision* (hallucination); saying “no” to present ones lowers *recall* (omission). The *yes-ratio* is the share of “yes” answers, and the *AUROC* of the model’s probability of “yes” measures whether present objects are ranked above absent ones, independently of where the yes/no threshold lies. Finally, *blind perplexity*, the perplexity of reference captions given a text-only prompt with no image, measures the health of the language model on its own.

Hypotheses. Each hypothesis predicts a distinct pattern:

- **H1, Lexical Fallback:** CHAIR_i and yes-ratio rise, precision falls, coverage is stable.
- **H2, Omission:** coverage (or recall) falls without a rise in CHAIR_i .
- **H0, no grounding-specific loss:** grounding changes only in step with blind perplexity.

POPE F1 and CHAIR alone cannot tell these apart: opposite changes in precision and recall can leave F1 flat, and omission lowers CHAIR.

Research questions. **RQ1:** Under deployed quantizers, does grounding change beyond what POPE F1 shows, and in which direction? **RQ2:** Is the change visual and quantization-specific, or a generic effect of weight error? **RQ3:** Does it enter at image or at text positions? **RQ4:** Can a training-free correction placed there remove it?

4 Experimental Setup

Model and data. LLaVA-1.5-7B (Liu et al., 2024a) encodes an image with CLIP ViT-L/14 into 576 *image tokens*, which a Vicuna-7B decoder reads together with the *text tokens* of the prompt and of its own answer. We quantize only the decoder (224 linear layers, 6.48B parameters). A fixed split defines a 500-image COCO val2014 evaluation pool and a disjoint 256-image calibration pool. POPE uses 100 images (1,800 questions); captions answer “Please describe this image in detail.” with greedy decoding (at most 128 tokens) on 100 images in Study 1 and 300 in Study 2.

Quantizers. $WbAa$ denotes b -bit weights and a -bit activations (Wb alone: 16-bit activations). We re-implemented GPTQ, AWQ, SmoothQuant and a QuaRot-style rotation on one simulated-quantization substrate, so that methods differ only in their algorithm; weights use groups of 128 (NF4: blocks of 64). Calibration uses 64 sequences of either WikiText-2 text (Merity et al., 2017) (*wiki*, the common default), COCO images with captions (*img+cap*), or the captions alone (*cap*). At 3 bits our AWQ and GPTQ close 35–36% of RTN’s WikiText-2 perplexity gap to FP16, at least the share reported for Llama-2-7B, and our simulated NF4 matches the real bitsandbytes kernels (Appendix A).

Diagnostic interventions. Two interventions answer RQ2 and RQ3. *Position-selective quantization:* in every linear layer, outputs at image-token positions are computed with one set of weights (FP16 or quantized) and outputs at text positions with the other, so damage can be attributed to one position type. *Matched noise:* the rounding error of RTN is replaced, in every group of 128 weights, by Gaussian noise of exactly the same energy; if noise harms the model in the same way, the harm comes from the size of the error and not from the structure of rounding.

Statistics. All changes are paired: quantized minus FP16 on the same images and questions.

We give 95% bootstrap intervals that resample *images* (2,000 resamples; Efron and Tibshirani, 1994), since questions about one image are correlated, and mark changes whose interval excludes zero with *. For families of tests we report Benjamini–Hochberg adjusted q -values (Benjamini and Hochberg, 1995).

5 Results: Diagnosing the Failure

5.1 Activations break on the language side

We first swept plain RTN from W4A8 down to W4A4 (Appendix B). W4A8 shows no significant change, and at W4A4 the model collapses (71% of POPE answers over the three splits are unparseable). Position-selective activation quantization locates the collapse: 4-bit activations at image positions only (text at 8 bits) leave POPE unchanged, whereas 4-bit activations at text positions only reproduce the collapse, and the collapsed model fails even without an image (blind perplexity 143 vs. 15.9). Text tokens carry the outlier “massive” activations (Sun et al., 2024), up to $|x| = 1564$ in the second layer. This confirms by direct intervention MBQ’s gradient-based finding that language tokens are the sensitive ones (Li et al., 2025). Notably, the first damage in the sweep is already omission: at W4A5, POPE recall falls from 0.87 to 0.64 while precision rises from 0.85 to 0.96, and captions shrink from 88 to 70 words.

5.2 Study 1: deployed quantizers cause omission

Study 1 compares 16 settings with FP16 (Figure 1): NF4; RTN, AWQ and GPTQ with 4- and 3-bit weights (at 3 bits, with each calibration source); RTN and SmoothQuant at W8A8; SmoothQuant at W4A8; and rotation at W4A4 with RTN or GPTQ weights.

POPE F1 is blind to the change. F1 moves by at most 0.02 in every setting. Underneath, the yes-ratio shifts in both directions (+6.1 points for W3 AWQ with wiki calibration, −3.1 for W3 RTN), with precision and recall moving in opposite directions: POPE registers a moved decision threshold, not a change in what the model sees.

Coverage falls in every setting. Coverage is below FP16 in all 16 settings (sign test, $p = 3 \cdot 10^{-5}$): by about 0.025 at W8A8 and W4A8, 0.011–0.045 with 4-bit weights, and 0.038–0.083 with 3-bit

weights; 9 of the 16 intervals exclude zero. Captions stay as long (about 92 words), and CHAIR_{*i*} rises significantly in none of the 11 weight-only settings. Quantized captions name fewer of the objects in the image without inventing more: the signature of H2. The only setting with a significant CHAIR_{*i*} rise, W4A4 rotation with RTN weights (+0.063), also has the most damaged language model (blind perplexity $1.20 \times$ FP16), so fallback appears only when the language model itself breaks.

5.3 Study 2: confirmation and controls

Study 2 re-tests six settings with three times as many captions (300; POPE unchanged) and adds the two interventions of §4, POPE AUROC, and a per-object analysis: for every object FP16’s caption names, does the quantized caption still name it? Table 1 gives the results.

Omission is robust. Coverage falls significantly in five of the six settings (−0.018 to −0.072) and CHAIR_{*i*} falls in four (−0.026 to −0.036). All 3-bit coverage drops survive multiplicity adjustment over the ten core settings ($q \leq 0.03$); the 4-bit GPTQ drop does not ($q = 0.06$). AUROC changes by at most 0.007 for the deployed quantizers, so they still rank present above absent objects; the POPE yes-ratio shifts are threshold moves. NF4, the default 4-bit format of the bitsandbytes library, shows no significant change on any measure.

Small objects are lost first. Among the deployed quantizers, large objects are dropped at 3–5% and small objects at 11–28%. Small objects (<1% of the image) are dropped significantly more often than the largest (>20%) in all 15 perturbed settings, including real-kernel NF4. Part of this is churn on marginal objects (FP16 itself names only 61% of small objects, and settings without any other significant change still drop 11–14% of them), but above this floor every additional unit of damage removes small objects first, up to 28% for W3 RTN.

The damage enters at text positions. With 3-bit weights used only at image positions, coverage and CHAIR_{*i*} stay at FP16 level (−0.008; small-object drops at the floor). With 3-bit weights only at text positions, the full loss appears: coverage −0.069 against −0.072 for the fully quantized model (paired difference +0.003, 95% CI [−0.013, 0.018]). The visual tokens survive 3-bit weights intact; what fails is the computation at text positions that turns visual information into words.

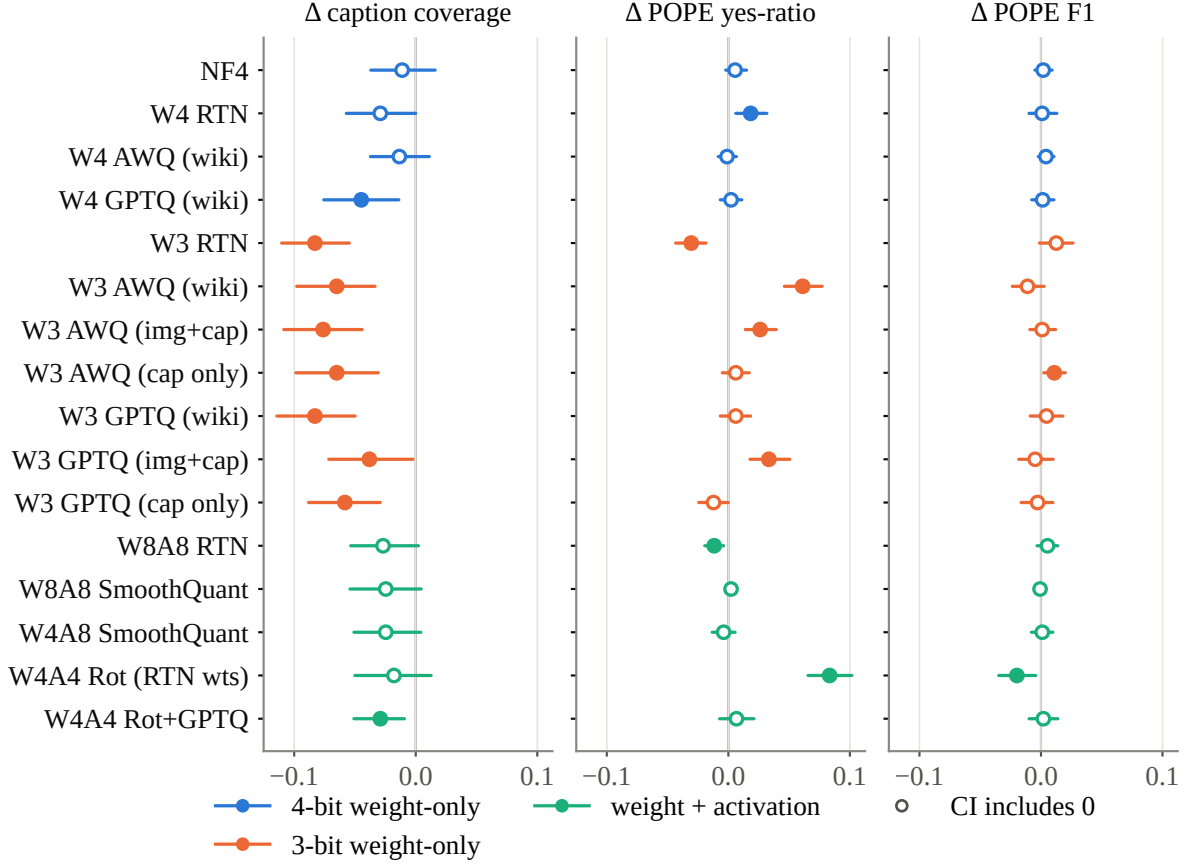


Figure 1: Study 1: change against FP16 in 16 deployed settings (100 POPE images, 1,800 questions; 100 caption images). Whiskers are paired 95% bootstrap intervals over images; filled markers exclude 0. FP16: coverage 0.774, yes-ratio 0.542, POPE F1 0.854. Coverage falls in all 16 settings while F1 stays within ± 0.02 .

This mirrors the activation result of §5.1 and fits evidence that hallucination in full-precision MLLMs arises in the language model rather than in visual encoding (Li et al., 2026).

Omission is a generic response to weight error. Random noise of RTN’s error energy also lowers coverage (-0.018 to -0.097 over three seeds at W3, bracketing RTN’s -0.072), but its effect on CHAIR_i and the yes-ratio changes sign from seed to seed. Losing objects is thus the decoder’s typical response to weight error of this size, whereas invented objects are not. Quantizers differ in how much of this error they leave: paired against W3 RTN, GPTQ recovers coverage ($+0.027$ with wiki calibration, $+0.048$ with img+cap; both significant). Image-and-caption calibration retains more coverage than WikiText calibration ($+0.021$, $p = 0.048$), a promising lead that does not yet survive multiplicity adjustment.

Quantized models attend less to the image. We measured, for every generated token, the share of

its attention that goes to the 576 image tokens, averaged over all layers and heads, on 100 POPE images and 100 captions (GPTQ calibrated on COCO captions; nucleus sampling for POPE). 4-bit GPTQ weights reduce it by 14% on POPE and 18% on captions, equally on all three POPE splits, and spread it more evenly over the image (Table 2). The drop comes from the weights: adding 8-bit activations to the same weights leaves it unchanged, and W8A8 changes it by only +2%. Within captions, names of invented objects receive less image attention than names of real ones at every precision (point-biserial $r = 0.19$ – 0.27 , all intervals above zero), so image attention tracks grounding word by word. Two cautions apply: POPE attention partly reflects the answer (“Yes, there is. . .” draws attention to the image), though 4-bit GPTQ attends less while answering “yes” slightly *more* often (59.6% vs. 58.2%); and on captions each setting attends while generating its own text. Attention is also more sensitive than behaviour: an 18% attention drop at 4 bits coexists with only a 1.8-point coverage loss for

Setting	change vs. FP16				objects dropped		blind
	Cov	CHAIR _i	yes-ratio	AUROC	small	large	PPL
FP16 (absolute values)	.744	.220	.542	–			15.6
NF4 (bitsandbytes kernels)	−.005	+.016	+.006	−.001	.11	.04	1.01×
W4A4 rot. + GPTQ (wiki)	−.002	−.001	+.007	−.001	.14	.03	1.06×
W4 GPTQ (wiki)	−.018*	−.026*	+.002	.000	.16	.03	1.01×
W3 GPTQ (img+cap)	−.025*	−.036*	+.033*	−.006*	.20	.03	1.13×
W3 AWQ (wiki)	−.042*	−.014	+.061*	−.004	.22	.04	1.10×
W3 GPTQ (wiki)	−.045*	−.028*	+.006	−.007*	.24	.04	1.09×
W3 RTN	−.072*	−.035*	−.031*	−.001	.28	.05	1.08×
<i>Matched noise: Gaussian weight noise with RTN’s per-group error energy</i>							
W4 noise, seed 0	−.027*	−.030*	+.034*	−.003*	.13	.05	1.09×
W4 noise, seed 1	−.018	−.019	+.001	.000	.17	.03	1.07×
W3 noise, seed 0	−.097*	−.046*	+.009	−.010*	.31	.08	1.14×
W3 noise, seed 1	−.069*	−.066*	−.081*	−.006	.25	.07	1.19×
W3 noise, seed 2	−.018	+.040*	+.041*	−.007	.18	.04	0.97×
<i>Position-selective quantization: W3 RTN weights at one position type, FP16 weights at the other</i>							
image positions only	−.008	−.006	−.017*	+.001	.13	.03	1.00×
text positions only	−.069*	−.040*	−.022*	−.002	.28	.05	1.08×

Table 1: Study 2 (300 caption images; 100 POPE images, 1,800 questions). Changes are quantized minus FP16 on the same items; *: paired 95% interval excludes 0. AUROC: of $P(\text{yes})$, pooled over POPE’s three splits (FP16 per split: .963, .951, .917). Objects dropped: of the annotated objects that FP16’s caption names, the share the setting’s caption no longer names, for objects covering under 1% (small) or over 5% (large) of the image; FP16 names 61% of small and 91% of large annotated objects. Blind PPL: relative to FP16 (it equals FP16 and W3 RTN in the last two rows by construction, since the test has no image). Simulated NF4 gives the same POPE metrics as the real kernels and coverage within 0.001.

Setting	Δ image attention		entropy (bits)
	POPE	captions	
FP16	–	–	5.49
W8A8 RTN	+2.0%	+3.9%	5.15
W4 GPTQ (cap)	−14.0%	−18.3%	6.29
W4A8 GPTQ (cap)	−12.9%	−17.5%	6.00

Table 2: Share of attention that generated tokens pay to the 576 image tokens: paired change against FP16 (FP16 share: 0.103 on POPE, 0.100 on captions; 100 POPE images, 100 captions). All 95% intervals exclude 0 and are within ± 1.4 points. Entropy: of attention over the image tokens, on captions (higher = more diffuse).

4-bit GPTQ (Table 1), so attention is an indicator of the failure rather than a measure of it.

Answers to RQ1–RQ3. Table 3 summarizes the verdicts. **RQ1:** deployed quantizers reduce grounding by up to 7.2 coverage points at 3 bits, a loss POPE F1 does not register, and the direction is

omission (H2): invented objects do not increase for any weight-only quantizer, and POPE yes-ratio shifts are threshold moves with AUROC unchanged. **RQ2:** the loss is visual (small objects first) and grounding-specific (H0 fails: among the 3-bit quantizers, GPTQ with img+cap has the highest blind perplexity, 1.13×, but the smallest coverage loss, −0.025), and it is a generic consequence of weight error rather than of rounding. **RQ3:** it enters at text positions, and the quantized model’s text tokens attend less to the image.

6 Results: Repairing the Failure

LoRAS-T. Because the damage enters at text positions, we correct the decoder exactly there. LoRAS-T (Low-Rank Activation Steering at Text positions) adds after every decoder layer ℓ a low-rank linear correction to the hidden state $h_\ell \in \mathbb{R}^{4096}$ of every text token, prompt and generated alike, and leaves image tokens untouched:

$$h_\ell \leftarrow h_\ell + h_\ell A_\ell B_\ell + b_\ell,$$

Hypothesis	Predicted signature	What we observe
H1: Lexical Fallback	$\text{CHAIR}_i \uparrow$, yes-ratio $\uparrow$, coverage stable	Not supported. CHAIR_i never rises significantly for weight-only quantizers; the yes-ratio moves both ways with AUROC flat. Fallback appears only when the language model itself breaks.
H2: Omission	coverage $\downarrow$ without $\text{CHAIR}_i \uparrow$	Supported. Coverage falls in 16 of 16 settings, by up to 7.2 points; small objects go first; the loss enters at text positions.
H0: no grounding-specific loss	grounding changes track blind perplexity	Rejected at 3 bits. Among the 3-bit quantizers, the one with the highest blind perplexity (GPTQ img+cap, $1.13\times$) has the smallest coverage loss.

Table 3: What each hypothesis predicts and what Studies 1 and 2 observe.

with $A_\ell \in \mathbb{R}^{4096 \times 64}$ and $B_\ell \in \mathbb{R}^{64 \times 4096}$. The correction is fitted to map the quantized model’s hidden states onto FP16’s on the same inputs, by reduced-rank ridge regression in closed form (Scetbon and Hensman, 2024; Fu et al., 2025): with centred quantized states X and residuals $R = H^{\text{FP16}} - H^Q$, we compute the ridge estimate $M = (X^\top X + \lambda I)^{-1} X^\top R$, take the top-64 eigenvectors V of $M^\top X^\top X M$, and set $A = MV$, $B = V^\top$. Layers are fitted in order, each with all earlier corrections active, so every correction sees the inputs it will see at inference. Calibration uses the 192 calibration-pool images with FP16’s own captions plus two yes/no object questions per image answered by FP16 (the questions name one annotated and one absent object); a few massive-activation tokens per sequence are excluded. Fitting takes 30 minutes on one A6000 GPU with no gradient steps, and LoRAS-T adds 16.9M parameters (34 MB, 0.26% of the decoder) and two rank-64 matrix products per layer and token.

LoRAS-T restores FP16-level grounding at 3 bits. Table 4 shows the result. On W3 RTN, LoRAS-T recovers 83% of the coverage loss ($+0.060^*$ paired against W3 RTN) and halves small-object drops, from 28% to 14%, the floor reached only by settings with no other measurable damage. The corrected model shows no significant difference from FP16 on any grounding measure: coverage, CHAIR_i , POPE F1, precision, recall, yes-ratio and AUROC (all POPE changes within 0.01); the only significant change is 1.9 more words per caption. CHAIR_i rises relative to W3 RTN ($+0.045^*$) but only back to the FP16 level: the correction restores FP16’s balance between naming and inventing rather than buying coverage with

invented objects. On every measure in Table 4, 3-bit RTN with LoRAS-T is at least as close to FP16 as 4-bit GPTQ, a quantizer that stores 0.8 GB more. On W3 GPTQ, LoRAS-T recovers 44% of the coverage loss ($+0.020^*$ against GPTQ) and cuts small-object drops from 24% to 15%. Because the correction never touches an image token, these results also confirm the attribution of §5.3 by intervention.

Calibration must include questions. Calibrated on captions alone, LoRAS-T already recovers 58% (RTN) and 62% (GPTQ) of the coverage loss, but on GPTQ it raised the POPE yes-ratio by 8.2 points while AUROC stayed at the GPTQ level: ranking was intact, but the threshold for answering “yes” had moved, because the correction had never seen a question prompt and the POPE answer is decided at the last prompt token. Adding yes/no questions answered by FP16 to calibration removes the shift for both quantizers (yes-ratio within 0.6 points of FP16), at a small coverage cost on GPTQ.

7 Plan for the Remaining Term

Table 5 gives the timeline. Three experiments carry the final claims: confirming LoRAS-T on images not used so far and across calibration seeds; correcting all positions versus text positions only, which tests whether the placement, not the added parameters, is what helps; and a full-rank closed-form correction (Fu et al., 2025), the closest prior method. We will also replicate the diagnosis on a second model and combine LoRAS-T with entropy-gated attention biasing, which raises attention from text tokens to the image when the model is uncertain.

Setting	change vs. FP16				blind	objects dropped	
	Cov	CHAIR _i	yes-ratio	AUROC	PPL	small	all
W3 RTN	−.072*	−.035*	−.031*	−.001	1.08×	.28	.16
+ LoRAS-T (captions only)	−.030*	+ .003	−.023*	−.001	0.99×	.17	.10
+ LoRAS-T	− .012	+ .009	+ .006	.000	0.96×	.14	.08
W3 GPTQ (wiki)	−.045*	−.028*	+ .006	−.007*	1.09×	.24	.13
+ LoRAS-T (captions only)	−.018	+ .012	+ .082*	−.006*	1.05×	.13	.08
+ LoRAS-T	−.025*	+ .008	+ .004	−.003	1.02×	.15	.09
W4 GPTQ (wiki), for reference	−.018*	−.026*	+ .002	.000	1.01×	.16	.08

Table 4: LoRAS-T on the Study 2 evaluation set (300 captions, 1,800 POPE questions; same FP16 reference as Table 1). *: paired 95% interval excludes 0. Objects dropped: for small objects (<1% of the image) and over all sizes. “Captions only”: calibration without the yes/no questions.

Dates	Work
Oct 3–9	Confirm LoRAS-T on the 200 unused evaluation images and 500 POPE images, with three calibration seeds. Attention on the Study 2 setup with teacher-forced FP16 captions (identical tokens for all settings), for W3, text-only W3 and LoRAS-T.
Oct 10–16	Second model (Qwen2.5-VL-7B; Bai et al., 2025): Study 2 core settings and LoRAS-T. Ablations: rank, layers, calibration size, and correction at all positions vs. text positions only.
Oct 17–23	Baselines: full-rank closed-form correction (Fu et al., 2025), img+cap calibration, MBQ, VCD. Entropy-gated attention biasing (A-CAB) on top of LoRAS-T; AWQ; general benchmarks (TextVQA, MME); real-kernel latency.
Oct 24–31	Final report, code release, viva.

Table 5: Timeline to the final submission (October 31).

8 Conclusion

Quantization does not make LLaVA-1.5 hallucinate; it makes it see less. Deployed quantizers remove present objects from captions, small objects first, a loss that POPE F1 does not register. The visual tokens survive quantization; the language side that reads them out does not. Correcting the decoder only at text positions, with a 34 MB closed-form correction, brings 3-bit RTN back to full-precision grounding.

Limitations

All quantizers except bitsandbytes NF4 are simulated, and our re-implementations differ from reference code in calibration data, grid convention and rotation scheme; the KV cache is not quantized. Results cover one model, 100 POPE and 100–300 caption images (60 and 40 for the RTN sweep), and greedy decoding except for the attention measurements. Intervals are unadjusted except where

stated, and the noise control has two or three draws per bit width. CHAIR and coverage rely on COCO annotations and a synonym list. The attention measurements use a different calibration and decoding setup from the other experiments. LoRAS-T was calibrated once per setting, its question-augmented calibration was designed after inspecting results of the caption-only version, and “no significant difference” at $n=300$ does not establish equivalence (coverage intervals are about ± 0.017). Being distilled from FP16, it also inherits FP16’s own errors.

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A Validation of the Quantizers

Table 6 checks the re-implementations. Calibrated on WikiText-2, AWQ and GPTQ close 35–36% of RTN’s 3-bit gap, matching or exceeding the published Llama-2-7B shares (at 4 bits our AWQ closes less, 10% against 50%); rotation reduces W4A4 perplexity from 1001 to 10.6, and to 8.5 with GPTQ weights calibrated on img+cap (the main experiments use wiki calibration); SmoothQuant (img+cap) W8A8 is within 0.04 of FP16. Calibrated on images with captions, GPTQ is worse than RTN on WikiText-2, because 88% of its calibration tokens are image tokens; calibration domain matters, which is why we report each source. Simulated NF4 gives the same POPE metrics as the real bitsandbytes kernels and coverage within 0.001 (Table 1).

Setting	calib.	PPL	closed	publ.
FP16	–	7.27		
W4 RTN	–	7.50	–	
W4 AWQ	wiki	7.47	10%	50%
W4 GPTQ	wiki	7.45	19%	15%
NF4	–	7.39		
W3 RTN	–	9.14	–	
W3 AWQ	wiki	8.47	36%	35%
W3 GPTQ	wiki	8.49	35%	19%
W3 AWQ	img+cap	8.75	21%	
W3 GPTQ	img+cap	9.87	–39%	
W8A8 RTN	–	7.30		
W8A8 SQ	img+cap	7.31		
W4A4 RTN	–	1001	–	
W4A4 rotation	–	10.64		
W4A4 rot.+GPTQ	img+cap	8.53		

Table 6: WikiText-2 perplexity of LLaVA’s decoder (40 windows of 2,048 tokens). Closed: share of RTN’s gap to FP16 that the method removes; publ.: the same share for Llama-2-7B in the AWQ paper (Lin et al., 2024, Table 4).

B Round-to-Nearest Precision Sweep

Table 7 gives the RTN sweep of §5.1.

RTN setting	F1	yes	Cov	PPL _∅
FP16	.860	.51	.76	15.9
W4A8	.869	.52	.76	15.7
W4A6	.868	.49	.72	17.0
W4A5	.767*	.33	.66	19.1
W4A4	.272*	.15	.01	143
A4 image, A8 text	.849	.48	.73	–
A4 text, A8 image	.124*	.08	.01	–

Table 7: RTN sweep (60 POPE images, random split; 40 caption images). PPL_∅: blind perplexity (on a 40-caption probe, hence a different FP16 value from Table 1). *: interval of the difference to FP16 excludes 0. The last two rows quantize activations to 4 bits at one position type only. With 4-bit activations at image positions only, CHAIR_i rises slightly (+0.056* on 40 captions), the only hallucination-like signal in this sweep.